

ENGINEERING PHYSICS · UNIT V

Semiconductors

Complete Formula & Concept Revision Sheet

LEVEL

B.Tech 1st Year

SEMESTER

1st Semester

UNIT

Unit V

PURPOSE

Last-Day Revision

TOPICS COVERED

- ▶ Energy Band Theory
- ▶ Intrinsic Semiconductors
- ▶ Density of Carriers
- ▶ Fermi Level
- ▶ Einstein's Relation
- ▶ Classification of Solids
- ▶ Extrinsic Semiconductors
- ▶ Electrical Conductivity
- ▶ Drift & Diffusion Currents
- ▶ Hall Effect

Formulas · Concepts · Flashcards · Exam Tips · Quick Revision



1 FORMATION OF ENERGY BANDS

Key Concepts

- Isolated atoms → discrete energy levels
- Atoms in crystal → levels split into **energy bands**
- Pauli exclusion principle → N atoms → N closely spaced levels → band

BAND DEFINITIONS

Valence Band (VB) Highest filled band at 0 K; electrons bound to atoms

Conduction Band (CB) Next higher band; free electrons reside here

Band Gap E_g Forbidden energy region between VB and CB

Band Diagram (Conceptual)

Conduction Band (CB) — free electrons

----- E_g = Band Gap -----

Valence Band (VB) — bound electrons

↑ E_F = Fermi level position depends on semiconductor type

BAND GAP ENERGY

$E_g = E_C - E_V$

2 CLASSIFICATION OF CRYSTALLINE SOLIDS

PROPERTY	CONDUCTORS (METALS)	SEMICONDUCTORS	INSULATORS
Band Gap (E_g)	≈ 0 eV (overlapping bands)	0.1 – 3.5 eV	> 5 eV
Conductivity (σ)	$10^6 - 10^8$ S/m	$10^{-4} - 10^4$ S/m	< 10^{-10} S/m
Resistivity (ρ)	Very Low	Intermediate	Very High
Temp Effect on σ	Decreases with ↑T	Increases with ↑T	Increases with ↑T (slight)
Examples	Cu, Ag, Al	Si (1.1 eV), Ge (0.67 eV)	SiO ₂ , Diamond (5.5 eV)
CB at 0 K	Partially filled	Empty	Empty

3 INTRINSIC SEMICONDUCTORS

DEFINITION

Pure semiconductor with no impurities. Thermal energy alone generates electron-hole pairs. Every electron promoted to CB leaves behind a hole in VB.

KEY RELATIONS

- For every e^- in CB → 1 hole in VB
- $n = p = n_i$ (intrinsic carrier concentration)

INTRINSIC CARRIER CONCENTRATION

$n_i = \sqrt{(N_C \cdot N_V) \cdot \exp(-E_g/2kT)}$

EFFECTIVE DENSITY OF STATES

$N_C = 2(2\pi m^*_e kT/h^2)^{3/2}$

$N_V = 2(2\pi m^*_h kT/h^2)^{3/2}$

FERMI LEVEL POSITION (INTRINSIC)

$E_F = (E_C + E_V)/2 + (3kT/4) \cdot \ln(m^*_h/m^*_e)$

NOTE

If $m^*_h \approx m^*_e$, then $E_F \approx \text{midgap } (E_C + E_V)/2$

4 EXTRINSIC SEMICONDUCTORS

N-TYPE SEMICONDUCTOR

Doped with **pentavalent** (Group V) impurity: P, As, Sb
Impurity donates extra electron → **donor atoms**
Donor level E_D lies just below E_C

P-TYPE SEMICONDUCTOR

Doped with **trivalent** (Group III) impurity: B, Al, Ga
Impurity accepts electron → creates hole → **acceptor atoms**
Acceptor level E_A lies just above E_V

N-TYPE CARRIER CONCENTRATIONS

$n \approx N_D$ (majority carriers = electrons)
 $p = n_i^2 / N_D$ (minority carriers = holes)

P-TYPE CARRIER CONCENTRATIONS

$p \approx N_A$ (majority carriers = holes)
 $n = n_i^2 / N_A$ (minority carriers = electrons)

FERMI LEVEL – N-TYPE

$E_F = E_C - kT \cdot \ln(N_C/N_D)$

FERMI LEVEL – P-TYPE

$E_F = E_V + kT \cdot \ln(N_V/N_A)$

FERMI LEVEL SHIFT

E_F shifts **towards Conduction Band**
More doping → E_F closer to E_C

FERMI LEVEL SHIFT

E_F shifts **towards Valence Band**
More doping → E_F closer to E_V

PROPERTY	INTRINSIC	N-TYPE	P-TYPE
Impurity	None (pure)	Pentavalent (Group V)	Trivalent (Group III)
Majority Carrier	e⁻ = holes	Electrons	Holes
Minority Carrier	–	Holes	Electrons
Fermi Level	Midgap	Near E_C	Near E_V
Conductivity	$\sigma = qn_i(\mu_n + \mu_p)$	$\sigma \approx qN_D\mu_n$	$\sigma \approx qN_A\mu_p$

5 DENSITY OF CHARGE CARRIERS

ELECTRON CONCENTRATION (GENERAL)

$n = N_C \cdot \exp[-(E_C - E_F)/kT]$

MASS ACTION LAW (CRITICAL)

$n \cdot p = n_i^2$

HOLE CONCENTRATION (GENERAL)

$p = N_V \cdot \exp[-(E_F - E_V)/kT]$

TEMPERATURE DEPENDENCE

$n_i \propto T^{3/2} \cdot \exp(-E_g/2kT)$
 $\uparrow T \rightarrow \uparrow n_i \rightarrow \uparrow \sigma$ (unlike metals!)

6 ELECTRICAL CONDUCTIVITY

GENERAL CONDUCTIVITY FORMULA

$\sigma = q(n\mu_n + p\mu_p)$

INTRINSIC (N = P = N_i)

$\sigma_i = qn_i(\mu_n + \mu_p)$

RESISTIVITY

$\rho = 1/\sigma = 1/[q(n\mu_n + p\mu_p)]$

Mobility

μ_n Electron mobility ($\text{cm}^2/\text{V}\cdot\text{s}$). For Si: ≈ 1350
 μ_p Hole mobility ($\text{cm}^2/\text{V}\cdot\text{s}$). For Si: ≈ 480
 $\mu_n > \mu_p$ Electrons always faster than holes

TEMPERATURE EFFECT ON MOBILITY

$\mu \propto T^{-3/2}$ (lattice scattering dominant at high T)
 $\mu \propto T^{+3/2}$ (impurity scattering at low T)
Overall: **μ decreases** as T increases

7 FERMİ LEVEL

DEFINITION
The energy level at which the probability of finding an electron is exactly **50%** (0.5).
Defined by the Fermi-Dirac distribution function:

FERMI-DIRAC DISTRIBUTION
$$f(E) = 1 / [1 + \exp((E-E_F)/kT)]$$

AT $E = E_F$: $F(E_F) = 0.5$ ALWAYS

SEMICONDUCTOR TYPE	FERMI LEVEL POSITION
Intrinsic ($m^*_e \approx m^*_h$)	$E_F \approx (E_C + E_V)/2$ (midgap)
n-type lightly doped	Just below E_C
n-type heavily doped	May cross E_C (degenerate)
p-type lightly doped	Just above E_V
p-type heavily doped	May go below E_V (degenerate)

TEMPERATURE EFFECT ON E_F
At very high T: all extrinsic semiconductors → intrinsic behavior → E_F → midgap

8 DRIFT CURRENT

CONCEPT
Motion of charge carriers due to applied **electric field E** .
Electrons drift opposite to E ; holes drift along E .

DRIFT VELOCITY
$$v_d = \mu E \quad (\text{m/s or cm/s})$$

DRIFT CURRENT DENSITY (ELECTRONS)
$$J_{n,\text{drift}} = nq\mu_n E$$

DRIFT CURRENT DENSITY (HOLES)
$$J_{p,\text{drift}} = p q \mu_p E$$

TOTAL DRIFT CURRENT DENSITY
$$J_{\text{drift}} = q(n\mu_n + p\mu_p)E = \sigma E$$

OHM'S LAW FORM
$$J = \sigma E = E/\rho$$

- KEY NOTES**
- Drift current exists only when $E \neq 0$
 - Both electrons and holes contribute **in same direction** to conventional current
 - $v_d \propto E$ (valid for low fields only)

9 DIFFUSION CURRENT

CONCEPT
Motion of carriers from **high concentration** → **low concentration** region.
Driven by concentration gradient — no electric field needed.

ELECTRON DIFFUSION CURRENT DENSITY
$$J_{n,\text{diff}} = qD_n (dn/dx)$$

HOLE DIFFUSION CURRENT DENSITY
$$J_{p,\text{diff}} = -qD_p (dp/dx)$$

TOTAL CURRENT DENSITY (DRIFT + DIFFUSION)
$$J_n = qn\mu_n E + qD_n (dn/dx)$$
$$J_p = qp\mu_p E - qD_p (dp/dx)$$

FEATURE	DRIFT	DIFFUSION
Driving force	Electric field E	Conc. gradient dn/dx
Formula	$J = qn\mu E$	$J = qD (dn/dx)$
Coeff used	Mobility μ	Diffusion coeff D
Direction	Along/opposite E	High→Low conc.

10 EINSTEIN'S RELATION

THE RELATION (VERY IMPORTANT)
Links diffusivity and mobility — two different transport phenomena.

- KEY POINTS**
- $D/\mu = kT/q = V_T$ (thermal voltage)

EINSTEIN'S RELATION

$$D_n/\mu_n = D_p/\mu_p = kT/q$$

THERMAL VOLTAGE

$$V_T = kT/q \approx 0.026 \text{ V at } 300 \text{ K}$$

- At 300 K: $kT/q \approx 26 \text{ mV}$

- Tells us: higher mobility $\rightarrow$ higher diffusivity
- Valid for non-degenerate semiconductors

USEFUL FORMS

$$D_n = \mu_n(kT/q) \quad | \quad D_p = \mu_p(kT/q)$$

11 HALL EFFECT

CONCEPT

When current-carrying conductor/semiconductor is placed in a perpendicular magnetic field **B**, a transverse electric field (Hall field) is developed perpendicular to both **J** and **B**. This is the **Hall Effect**.

- PHYSICAL SETUP**
- Current I flows in x-direction
 - Magnetic field B in z-direction
 - Hall voltage V_H develops in y-direction
 - Lorentz force: $F = qv \times B$

HALL VOLTAGE

$$V_H = (IB)/(nqd) = R_H \cdot (IB/d)$$

HALL COEFFICIENT

$$R_H = 1/(nq) \quad [\text{for n-type: } -1/(nq)]$$
$$R_H = 1/(pq) \quad [\text{for p-type: } +1/(pq)]$$

HALL FIELD

$$E_H = V_H/d = R_H \cdot J \cdot B$$

CARRIER CONCENTRATION FROM HALL

$$n = IB / (qV_H \cdot d) = 1/(qR_H)$$

MOBILITY FROM HALL MEASUREMENTS

$$\mu_H = |R_H| \cdot \sigma = |R_H| / \rho$$

QUANTITY	N-TYPE	P-TYPE
Majority carriers	Electrons	Holes
R_H sign	Negative (-)	Positive (+)
Hall field direction	-y direction	+y direction
V_H polarity	Bottom face -ve	Bottom face +ve

- APPLICATIONS OF HALL EFFECT**
- ✓ Determine carrier type (n or p)

✓ Measure carrier concentration (n or p)

✓ Calculate mobility (μ)
- ✓ Measure magnetic field strength

✓ Hall sensors / Hall probes

✓ Characterize semiconductor materials

★ IMPORTANT SYMBOLS & UNITS

SYMBOL	QUANTITY	UNIT	TYPICAL VALUE (SI AT 300K)
n_i	Intrinsic carrier concentration	m^{-3} or cm^{-3}	$1.5 \times 10^{10} \text{ cm}^{-3}$
μ_n	Electron mobility	$cm^2/V \cdot s$	$1350 \text{ cm}^2/V \cdot s$
μ_p	Hole mobility	$cm^2/V \cdot s$	$480 \text{ cm}^2/V \cdot s$
E_g	Band gap (Si)	eV	1.12 eV
E_g	Band gap (Ge)	eV	0.67 eV
kT/q	Thermal voltage at 300K	V	0.026 V (26 mV)
q	Electron charge	C	$1.6 \times 10^{-19} \text{ C}$
k	Boltzmann constant	J/K	$1.38 \times 10^{-23} \text{ J/K}$



FLASHCARDS (25 CARDS)

Q: What is the band gap?

A: Energy difference between valence band (E_V) and conduction band (E_C). $E_g = E_C - E_V$

Q: What is Einstein's relation?

A: $D/\mu = kT/q = V_T \approx 0.026 \text{ V at } 300 \text{ K}$

Q: Where is Fermi level in n-type?

A: Shifts towards conduction band E_C . Heavier doping $\rightarrow$ closer to E_C

Q: What causes drift current?

A: Applied electric field E . Carriers move under Lorentz force. $J_{\text{drift}} = q(n\mu_n + p\mu_p)E$ Q: Hall coefficient R_H formula?A: $R_H = 1/(nq)$ for n-type (negative); $R_H = 1/(pq)$ for p-type (positive)

Q: Electrical conductivity formula?

A: $\sigma = q(n\mu_n + p\mu_p)$ (S/m or $\Omega^{-1}\text{m}^{-1}$)

Q: What is acceptor impurity?

A: Trivalent atom (B, Al, Ga) added to Si. Accepts electron, creates hole $\rightarrow$ p-type.

Q: Majority carriers in n-type?

A: Electrons. Concentration $\approx N_D$ (donor concentration)

Q: Effect of temperature on intrinsic semiconductor conductivity?

A: σ increases with T (more e-h pairs generated). Opposite to metals!

Q: Band gap of Silicon?

A: $E_g(\text{Si}) = 1.12 \text{ eV at } 300 \text{ K}$

Q: Hall mobility formula?

A: $\mu_H = |R_H| \cdot \sigma = |R_H|/\rho$ Q: What does N_C represent?A: Effective density of states in conduction band. $N_C = 2(2\pi m_e^* kT/h^2)^{3/2}$

Q: What is the mass action law?

A: $n \cdot p = n_i^2$ (valid for both intrinsic and extrinsic at thermal equilibrium)

Q: Where is Fermi level in intrinsic semiconductor?

A: At midgap: $E_F \approx (E_C + E_V)/2$ (when $m_e^* \approx m_h^*$)

Q: Where is Fermi level in p-type?

A: Shifts towards valence band E_V . Heavier doping $\rightarrow$ closer to E_V

Q: What causes diffusion current?

A: Concentration gradient (dn/dx or dp/dx). No electric field needed.

Q: How to find carrier type using Hall Effect?

A: If R_H is negative $\rightarrow$ n-type. If R_H is positive $\rightarrow$ p-type.

Q: What is donor impurity?

A: Pentavalent atom (P, As, Sb) added to Si. Donates extra electron $\rightarrow$ n-type.Q: Thermal voltage V_T at 300 K?A: $V_T = kT/q \approx 26 \text{ mV} = 0.026 \text{ V at room temperature}$

Q: Minority carriers in n-type?

A: Holes. Concentration $= n_i^2/N_D$

Q: Which is faster: electron or hole?

A: Electron. $\mu_n > \mu_p$. For Si: $\mu_n \approx 1350$, $\mu_p \approx 480 \text{ cm}^2/\text{V}\cdot\text{s}$

Q: Band gap of Germanium?

A: $E_g(\text{Ge}) = 0.67 \text{ eV at } 300 \text{ K}$ Q: Fermi-Dirac probability at $E = E_F$?A: $f(E_F) = 1/(1+1) = 0.5 \rightarrow$ exactly 50% probability of occupation

Q: What determines band gap category (conductor/SC/insulator)?

A: Conductors: $E_g \approx 0$. Semiconductors: $0.1\text{--}3.5 \text{ eV}$. Insulators: $>5 \text{ eV}$

Q: Hall voltage formula?

A: $V_H = IB/(nqd)$ where d = thickness of sample



COMMON EXAM MISTAKES & TIPS



Fermi Level Shift Direction: n-type $\rightarrow$ towards E_C (conduction band). p-type $\rightarrow$ towards E_V (valence band). Many students swap these. Remember: *n goes up, p goes down* on energy diagram.



Forgetting $np = n_i^2$: This law applies at *thermal equilibrium*. Both intrinsic and extrinsic follow this. Always use it to find minority carriers: $p = n_i^2/N_D$ for n-type, $n = n_i^2/N_A$ for p-type.



Wrong Units: σ in S/m (not $\Omega\cdot m$). $\rho = 1/\sigma$ in $\Omega\cdot m$. Mobility μ in $cm^2/V\cdot s$. Concentration n in cm^{-3} . Band gap E_g in eV. Current density J in A/m^2 .



Mixing Drift and Diffusion: Drift needs electric field ($E \neq 0$). Diffusion needs concentration gradient ($dn/dx \neq 0$). Total current = drift + diffusion. Don't confuse their driving forces.



Hall Coefficient Sign Error: R_H is **NEGATIVE for n-type** (electrons). R_H is **POSITIVE for p-type** (holes). The sign tells you the carrier type — don't drop the negative sign!



Temperature Dependence: In metals: σ decreases with T . In semiconductors: σ increases with T (because more carriers generated). This is because more e-h pairs are generated at high T in semiconductors.



Intrinsic n_i Formula: n_i depends on $\exp(-E_g/2kT)$, not $\exp(-E_g/kT)$. The factor of 2 in denominator comes from taking square root of n-p product.



Einstein Relation: $D/\mu = kT/q$ (not kT , not k/qT). At 300 K, $kT/q = 26 \text{ mV} \approx 0.026 \text{ V}$. This is the thermal voltage V_T .



Electron vs Hole Mobility: Always $\mu_n > \mu_p$. Electrons are more mobile. For Si: $\mu_n = 1350 \text{ cm}^2/V\cdot s$, $\mu_p = 480 \text{ cm}^2/V\cdot s$. Never say $\mu_p > \mu_n$!



Diffusion current hole sign: $J_{p,diff} = -qD_p(dp/dx)$. Note the negative sign for holes! For electrons: $J_{n,diff} = +qD_n(dn/dx)$. Sign convention is crucial.



QUICK REVISION — ALL FORMULAS AT A GLANCE

INTRINSIC CARRIERS

$$n = p = n_i$$

$n \cdot p = n_i^2$ (mass action)

CONDUCTIVITY

$$\sigma = q(n\mu_n + p\mu_p)$$

$$\rho = 1/\sigma$$

HALL COEFFICIENT

$$R_H = \pm 1/(nq)$$

-ve n-type, +ve p-type

CARRIER CONCENTRATION

$$n_i = \sqrt{N_c N_v} \cdot e^{(-E_g/2kT)}$$

Temp dependent

DRIFT VELOCITY

$$vd = \mu E$$

$$J = nqv_d = \sigma E$$

HALL VOLTAGE

$$V_H = IB/(nqd)$$

d = sample thickness

N-TYPE (EXTRINSIC)

$$n \approx N_D$$

$$p = n_i^2/N_D$$

DIFFUSION CURRENT

$$J_n = qD_n(dn/dx)$$

$$J_p = -qD_p(dp/dx)$$

HALL MOBILITY

$$\mu_H = |R_H| \cdot \sigma$$

$$= |R_H|/\rho$$

P-TYPE (EXTRINSIC)

$$p \approx N_A$$

$$n = n_i^2/N_A$$

EINSTEIN RELATION

$$D/\mu = kT/q$$

$$\approx 0.026 \text{ V at 300 K}$$

FERMI-DIRAC

$$f(E) = 1/[1 + e^{((E-E_f)/kT)}]$$

$$f(E_F) = 0.5$$



All the best for your exam! | Remember: $np = n_i^2 \cdot D/\mu = kT/q \cdot R_H$ sign tells carrier type